A Distributor-Centric Blockchain Framework for Regulatory-Aware Pharmaceutical Supply Chain Verification

PHASE I REPORT

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BONAFIDE CERTIFICATE

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"This project addresses the following Sustainable Development Goal(s):
SDG GOAL 3 : GOOD HEALTH AND WELL BEING

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Abstract

Distributors are currently the most affected by fake, expired, recalled, and prohibited drugs in India's pharmaceutical supply chain. The project is dedicated to developing a Distributor-Centric, Regulatory-Aware Blockchain Enforcement System that is designed to offer verifiable and trustworthy verification for the distributor-pharmacy-patient value chain. The medicines will be assigned QR codes and a verification engine that examines the presence of an item, expiry date, regulatory status, ownership, and supply chain history. The regulatory state of the medicines will be stored in a regulatory database that holds information about the items, including the Central Drugs Standard Control Organization (CDSCO) details. The blockchain ledger will store essential transactional data related to the medicines, such as receipt, transfer, blocking, and dispensation. Smart contracts can ensure that prohibited, expired, faulty or blocked medicines are not transferred or dispensed. Pharmacies will be able to use the verification engine to check medicines before dispensing them to the patients. The patients, on the other hand, can use the QR code to access the status of the medicine. The system is expected that the project will establish an enhanced value chain for pharmaceutical products by focusing on counterfeits and questionable items. It will also provide a basis for regulatory blocking of problematic medicines and the ability to perform essential functions in the supply chain infrastructure.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
	LIST OF TABLES	viii
	LIST OF FIGURES	ix
	LIST OF ABBREVIATIONS	x
1.	**INTRODUCTION**	**9**
	1.1 GENERAL	9
	1.2 OBJECTIVE	10
	1.3 EXISTING SYSTEM	10
	1.4 PROPOSED SYSTEM	11
2.	**LITERATURE SURVEY**	**13**
	2.1 LITERATURE REVIEW	13
	2.2 COMPARISON AND DISCUSSION	20
	2.3 CONCLUSION	20
3.	**SYSTEM DESIGN**	**21**
	3.1 SYSTEM ARCHITECTURE	21
	3.2 SYSTEM REQUIREMENTS	25
	3.2.1 SOFTWARE REQUIREMENTS	26
	3.2.2 HARDWARE REQUIREMENTS	26
	3.2.3 DATASET REQUIREMENTS	27
4.	**PROJECT DESCRIPTION**	**28**
	4.1 METHODOLOGIES	28
	4.2 MODULES	28
	4.2.1 USER AUTHENTICATION AND ROLE MANAGEMENT	28
	4.2.2 MEDICINE REGISTRATION AND QR VERIFICATION	29
	4.2.3 DISTRIBUTOR AND PHARMACY MANAGEMENT	29

4.2.4 INVENTORY AND SUPPLY-CHAIN TRACKING 29
4.5 REGULATORY VERIFICATION 30
4.6 BLOCKCHAIN AND SMART CONTRACT MANAGEMENT 30
4.7 ALERT AND AUDIT MANAGEMENT 31
4.8 PATIENT VERIFICATION 31
5. IMPLEMENTATION AND RESULT DISCUSSION 32
5.1 IMPLEMENTATION RESULT 32
5.1.1 IMPLEMENTATION OF USER AUTHENTICATION AND ROLE-BASED ACCESS 32
5.1.2 IMPLEMENTATION OF MEDICINE AND BATCH MANAGEMENT 33
5.1.3 IMPLEMENTATION OF QR/BARCODE-BASED MEDICINE VERIFICATION 33
5.1.4 IMPLEMENTATION OF THE MEDICINE VERIFICATION ENGINE 34
5.1.5 IMPLEMENTATION OF BLOCKCHAIN-BASED SUPPLY-CHAIN TRACKING 35
5.1.6 IMPLEMENTATION OF DISTRIBUTOR AND PHARMACY OPERATIONS 36
5.1.7 IMPLEMENTATION OF PATIENT VERIFICATION 37
5.1.7 IMPLEMENTATION OF ALERT AND ENFORCEMENT MECHANISM 38
5.1.8 IMPLEMENTATION OF DATABASE MANAGEMENT 38
5.1.9 IMPLEMENTATION OF AUDIT AND SECURITY MECHANISMS 39
5.2 DISCUSSION 40

6. CONCLUSION AND FUTURE WORK 41

6.1 CONCLUSION 41

6.2 FUTURE WORK 43

REFERENCES 45

LIST OF TABLES

TABLE NO.	TABLE NAME	PAGE NO.
3.1	Software requirements of the proposed system	17

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1	Architecture diagram of the proposed verification system	22
3.2	Activity Diagram of the system	25
5.1	User authentication and role-based access	32
5.2	Medicine and batch management	33
5.3	QR/barcode-based medicine verification	34
5.4	Medicine verification and enforcement engine	35
5.5	Blockchain-based supply-chain tracking	36
5.6	Patient verification	37
5.7	Alert and enforcement mechanism	39

LIST OF ABBREVIATIONS

API	Application Programming Interface
CDSCO	Central Drugs Standard Control Organisation
DLT	Distributed Ledger Technology
EVM	Ethereum Virtual Machine
HD	High Definition
JWT	JSON Web Token
MVP	Minimum Viable Product
QR	Quick Response
RAM	Random Access Memory
REST	Representational State Transfer
SSD	Solid State Drive
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 GENERAL

The pharmaceutical supply chain is an essential part of the healthcare system. However, there are different types of questionable medicines, such as fake, expired, rejected, and unauthorized ones, which threaten the health of many people. A medicine goes through several stages, including production, distribution, storage, and delivery to a patient, and there is a risk that it will be diverted during one of these stages. In this regard, the tracking system should ensure the visibility of pharmaceutical products throughout the supply chain. Tracking systems for pharmaceuticals are mostly based on centralized databases and require extensive documentation; therefore, they are vulnerable to security breaches and tampering. Moreover, there is often a significant delay between the time of product recall or denial and the time of updating of the databases, and such information must be available in real-time.

The project aims to propose a Distributor-Centric, Regulatory-Aware Blockchain Enforcement System for Pharmaceutical Verification. It should create a blockchain that starts from the distributor, goes through pharmacies, and ends with the patient. Each package of pharmaceutical products has a unique QR code with the product, batch, or serial number. All stakeholders can scan the code and receive relevant information about the medicine.

The system comprises blockchain technology, QR-based identification, regulatory database, and smart contracts. Blockchain will be used to maintain immutable records of key supply-chain events, while conventional databases will hold detailed product and regulatory information. The regulatory component will classify medicines and batches as permitted, restricted, recalled, prohibited, expired, and unknown.

When a distributor receives a medicine, the QR code is scanned and verified. The system will check the details of the product, expiry date, regulatory status, ownership, and transaction history. If the medicine is invalid, expired, recalled, or prohibited, the system will prevent further transactions. Pharmacies will perform another verification step before accepting or dispensing medicines, and patients can scan the QR code to check the medicine details.

1.2 OBJECTIVE

The objective of this project is to design a secure, reliable pharmaceutical verification system to enable medicine authentication, supply-chain tracking, and regulatory enforcement. The system will detect counterfeit, expired, recalled, and prohibited medicines using QR-based identification and blockchain technology. The project will also ensure that medicine transaction records are tamper-proof, restrict further distribution of invalid, unauthorized products, and allow verification by distributors, pharmacies, and patients.

1.3 EXISTING SYSTEM

Verification and tracking systems currently in use in the pharmaceutical sector are based on conventional databases, QR

or barcode recognition, manual recording of business transactions, and proprietary information systems used by manufacturers and distributors. Typically, the records of products, batches, invoices, and transactions are maintained in databases belonging to manufacturers or distributors. At the time of receiving medicines, a distributor scans the QR code or uses the invoice, product details, expiry date, or barcode to verify the product. The same applies when pharmacies accept medicines from distributors and patients receive products from pharmacies. In most cases, patients are unable to view or verify medicine details. One issue with verification and tracking systems is the lack of centralized information control, leading to medicines having different records in different databases. Another problem is that information may change over time, and different parties may have different versions of the medicine's batch, expiry date, or status. Another problem is a reliance on centralized databases that may be changed or corrupted. Another limitation is that conventional verification systems do not offer full protection against unauthorized changes to previously made transactions. Finally, enforcing regulations using such a system is challenging, especially when recalling medicines or changing the status of a product. The updated status needs to be reflected in all databases, and there is a risk of some components of the system not updating correctly, leading to errors and allowing prohibited medicines to reach patients.

1.4 PROPOSEDSYSTEM

The proposed system is Distributor-Centric, Regulatory-Aware Blockchain Enforcement System for pharmaceutical

verification. The key difference from conventional verification and tracking systems is that the proposed solution also enforces regulations. A QR code on the medicine package contains a product, batch, or serial number. The distributor scans the QR code and uses the application to verify the product. Smart contracts ensure that only authorized transactions can occur. Product details are obtained from the regulatory database, which classifies medicines and batches into permitted, restricted, recalled, prohibited, expired, and unknown status. The system will use a blockchain to maintain an immutable record of key events in the medicine supply chain. The events include receipt of medicine by a distributor, transfer of medicine between distributors, blocking of medicine, and dispensing by a pharmacy. To optimize performance, the product and transaction data are stored in conventional databases (off-chain), with only the blockchain component being used to store essential information about the medicine and its transaction history. Smart contracts will be used to enforce regulations. For instance, if a medicine is recalled or prohibited, it cannot be transferred from a distributor to a pharmacy. A medicine with an expired date cannot be dispensed to a patient. Before any of these transactions can take place, the medicine must be verified, and the product must have a permitted status. The proposed system will combine QR-based medicine verification, off-chain conventional databases, regulatory compliance verification, blockchain transaction records, and smart contract enforcement of regulations.

CHAPTER 2
LITERATURE SURVEY

2.1 LITERATURE REVIEW

The pharmaceutical supply chain comprises manufacturers, distributors, wholesalers, pharmacies, regulatory bodies, and patients. It presents challenges with traceability, counterfeit products, inaccurate records, transparency, and establishing reliable product provenance. Given the direct impact on patient safety, the ability to determine the origin of a product is particularly important in the pharmaceutical sector. The conventional pharmaceutical supply-chain systems typically rely on centralized databases and organizational record-keeping, which can lead to fragmented information and limited visibility across organizational boundaries. Blockchain and distributed ledger technologies (DLT) have, therefore, attracted significant attention as potential enablers for pharmaceutical supply-chain traceability and transparency. Musamih et al. [1] proposed “A Blockchain-Based Approach for Drug Traceability in Healthcare Supply Chain”, which serves as the primary basis for the proposed work. The study introduces an Ethereum-based blockchain framework for pharmaceutical traceability, whereby smart contracts are utilized to facilitate supply-chain transactions and drug provenance. The blockchain is supplemented with decentralized, off-chain storage of information associated with pharmaceutical products while maintaining tamper-resistant records of transactions. The main aim is to improve the ability of stakeholders to verify the origin of medicines and address counterfeit-drug problems arising from inadequate transparency and data provenance in the

conventional pharmaceutical supply chains. The study demonstrates that the blockchain can provide a trustworthy record of pharmaceutical transactions across multiple organizational participants and enable drug authenticity and provenance verification. The work of Musamih et al. provides an important technological foundation for pharmaceutical traceability; however, its primary emphasis is on maintaining trustworthy provenance and tracking drug transactions. A significant extension can be made considering the current regulatory status of a medicine or batch in addition to its historical supply-chain information. A medicine may be genuine and have a valid blockchain history while subsequently becoming expired, recalled, restricted or prohibited by a regulatory authority. Therefore, verifying only the authenticity and provenance of a product may not be sufficient to determine whether it should currently be allowed to proceed through the supply chain. The proposed work addresses this limitation with the introduction of a regulatory-aware verification and enforcement layer which combines supply-chain history with expiry and regulatory information prior to allowing further distribution. Lau, Liu and Au [2] proposed a Blockchain-Based Supply Chain System for Traceability, Regulation and Anti-Counterfeiting, which specifically addresses the difficulties associated with regulation enforcement and counterfeit prevention in regulated industries such as pharmaceutical manufacturing. Their system incorporates blockchain-based traceability alongside digital identity management and quality monitoring and was designed using real-world logistics and supply-chain practices. The authors also report deployment with drug manufacturers and wholesalers and evaluate the efficiency

of the implementation. This work demonstrates that blockchain can be applied beyond simple record storage to provide a shared and auditable infrastructure for regulated supply-chain environments. Although the above study explicitly considers regulation, its contribution primarily focuses on incorporating regulatory considerations, identity management and quality monitoring into a blockchain supply-chain architecture. The proposed work builds on this concept by conceptualizing regulation status as a dynamic operational state, which directly influences whether a supply-chain transaction can occur. In the proposed system, a medicine or batch can have statuses such as permitted, restricted, recalled, prohibited, expired or unknown. When the regulatory state changes, the system can identify affected inventory and prevent further distributor-to-pharmacy transactions. Thus, regulation is not only represented as information for stakeholders but is incorporated into the decision-making and enforcement mechanism of the supply chain. Another important contribution to pharmaceutical blockchain research was presented by Agrawal et al. [3] with “A Robust Drug Recall Supply Chain Management System Using Hyperledger Blockchain Ecosystem.”. The study specifically focuses on the drug-recall problem and proposes a blockchain-enabled pharmaceutical supply chain capable of supporting both forward drug distribution and backward recall operations. Hyperledger Composer is utilized to implement the proposed system, while blockchain transactions provide immutable and transparent records. The authors also develop mathematical models to reduce the time and cost involved in supplying medicines and recalling defective products. Their real-time implementation demonstrates the feasibility of blockchain-based

recall management and improved transparency in pharmaceutical distribution. The recall-oriented approach has high relevance to the proposed system since it demonstrates the importance of identifying and recovering affected pharmaceutical products. However, recall management generally focuses on tracing affected products after a recall decision has been made. The proposed system aims to extend this process by directly connecting regulatory events with downstream transaction enforcement. When a batch is newly classified as recalled or prohibited, the proposed system can update its operational status, identify relevant inventory and prevent the batch from being transferred further. This changes the role of blockchain from supporting recall traceability to supporting active prevention of further distribution. Gomasta et al. [4] introduced PharmaChain: Blockchain-based Drug Supply Chain Provenance Verification System, a Hyperledger Fabric-based framework designed to enhance pharmaceutical provenance and prevent counterfeit medicines. The system utilizes both on-chain and off-chain storage to support secure and efficient transactions and employs smart contracts for establishing data provenance. The study also considers security mechanisms such as digital signatures, hashing and resistance against certain blockchain attacks. The authors emphasize the importance of end-to-end tracking and reliable provenance information for protecting the pharmaceutical supply chain from counterfeit products. PharmaChain demonstrates the increasing shift from conventional centralized pharmaceutical databases to distributed and permissioned blockchain infrastructures. Its utilization of Hyperledger Fabric is particularly relevant since pharmaceutical supply chains involve identifiable organizations

that may require controlled participation and access. However, provenance verification primarily answers whether the recorded history of a product is trustworthy. It does not by itself determine whether the product is currently permitted for distribution under changing regulatory conditions. The proposed system builds on the provenance concept by combining blockchain history with an external regulatory database and expiry verification engine. Overall, the existing literature demonstrates that blockchain can provide substantial benefits for pharmaceutical supply chains. Blockchain-based approaches have been investigated for drug traceability, counterfeit detection, provenance verification, supply-chain monitoring, identity management and recall management. The literature also demonstrates the utilization of different blockchain architectures including Ethereum-based public or semi-public models and permissioned frameworks such as Hyperledger Fabric. These studies collectively establish that distributed ledgers can enhance transparency and create tamper-resistant records across organizations that participate in pharmaceutical distribution. However, several limitations remain when the objective shifts from simply tracking a medicine to actively determining whether it should continue moving through the supply chain. Existing systems frequently emphasize historical provenance, authenticity and traceability. These properties are essential but do not necessarily indicate the current regulatory eligibility of a product. For example, a medicine can have a completely valid blockchain history and still become unsuitable for distribution because its batch has expired, or it has been recalled, prohibited by a regulatory authority or is being handled by an unauthorized participant. This presents a gap between supply-chain

traceability and regulatory-aware enforcement. The proposed work addresses this gap by presenting a Distributor-Centric, Regulatory-Aware Blockchain Enforcement System for Pharmaceutical Verification. Each medicine package or batch is associated with a unique QR-based identifier that can be scanned by distributors, pharmacies and patients. At the distributor level, the verification engine checks the medicine's identity, batch information, expiry status, ownership and blockchain supply-chain history. In parallel, a regulatory database maintains the current status of medicines or batches based on authoritative regulatory information. The resultant information is combined to determine whether a medicine is permitted to proceed through the supply chain. A key feature of the proposed approach is the separation of regulatory decision-making from blockchain storage. Regulatory information can change over time and, therefore, should not be treated as an immutable blockchain record. Instead, the current regulatory status is maintained in an off-chain database, while important supply-chain events such as receiving, transferring, blocking and dispensing are recorded on the blockchain. Smart contracts can then enforce predefined rules based on the verified state of the medicine. This hybrid architecture attempts to retain the immutability and auditability of blockchain while avoiding the limitations of storing frequently changing regulatory information directly on-chain. The proposed system also extends verification beyond distributors and pharmacies to the patient. A patient can scan the medicine's QR code and obtain a simplified verification result indicating whether the medicine is genuine, within its expiry period, permitted by the regulatory system and associated with a valid supply-chain history. If the

medicine has been recalled, prohibited or otherwise blocked, the system can communicate an appropriate warning. This provides an additional transparency layer wherein the patient does not need direct access to complex blockchain transactions but can benefit from the information recorded and verified by the underlying system. Therefore, the research gap identified from the existing literature is not the absence of blockchain-based pharmaceutical traceability. Rather, the gap lies in the integration of dynamic regulatory information with blockchain-based supply-chain enforcement. Existing research provides great foundations for the provenance, counterfeit detection, traceability and recall management but there remains scope for a unified framework wherein regulatory status, expiry, ownership and supply-chain history are evaluated simultaneously and translated into an actionable allow-or-block decision. The proposed system aims to bridge this gap by moving from passive pharmaceutical traceability to active regulatory-aware enforcement, and also provide accessible verification information to the patient. The efficacy of the proposed framework can subsequently be demonstrated using valid, counterfeit, expired, recalled, prohibited and unauthorized medicine scenarios, and with metrics such as verification latency, detection accuracy, blocking efficacy, transaction processing time and regulatory-status propagation time.

2.2 COMPARISON AND DISCUSSION

Existing blockchain-based pharmaceutical systems mainly focus on traceability, authentication, counterfeit detection, and recall management. They provide an immutable record of medicine movement but generally do not dynamically enforce changing

regulatory conditions. The proposed system extends these capabilities by introducing a regulatory-aware enforcement layer. It checks medicine authenticity, expiry, ownership, and current regulatory status before allowing transfer or dispensing. If a batch is recalled, prohibited, or expired, the system can automatically block further transactions and alert distributors and pharmacies. Thus, the key difference is that existing systems mainly answer “Where did the medicine come from?”, while the proposed system also answers “Is this medicine currently allowed to proceed?” This moves pharmaceutical blockchain from passive traceability toward active regulatory enforcement, with verification extended from distributor to pharmacy and finally to the patient.

2.3 CONCLUSION

The proposed system combines blockchain, QR-based verification, and regulatory monitoring to improve the safety and transparency of the pharmaceutical supply chain. It verifies medicine authenticity, expiry, ownership, and current regulatory status while maintaining a secure transaction history. The system can automatically block expired, recalled, prohibited, or unauthorized medicines from further distribution. Overall, it provides a reliable solution for end-to-end pharmaceutical verification from distributor to pharmacy and finally to the patient, moving beyond basic traceability toward active regulatory enforcement.

CHAPTER 3
SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The Seven Layers of the Proposed Verification System The proposed pharmaceutical verification system follows a 7-layer architecture, addressing the requirements for medicine verification, supply-chain traceability, regulation enforcement, and auditability. The architecture includes the users and actors, frontend applications, backend APIs, medicine verification and enforcement engine, PostgreSQL database with regulatory data, blockchain layer, and alert and audit layer. The system implements a hybrid architecture, storing the sensitive operational information in a PostgreSQL database while maintaining important supply-chain related transactions on an Ethereum compatible blockchain.

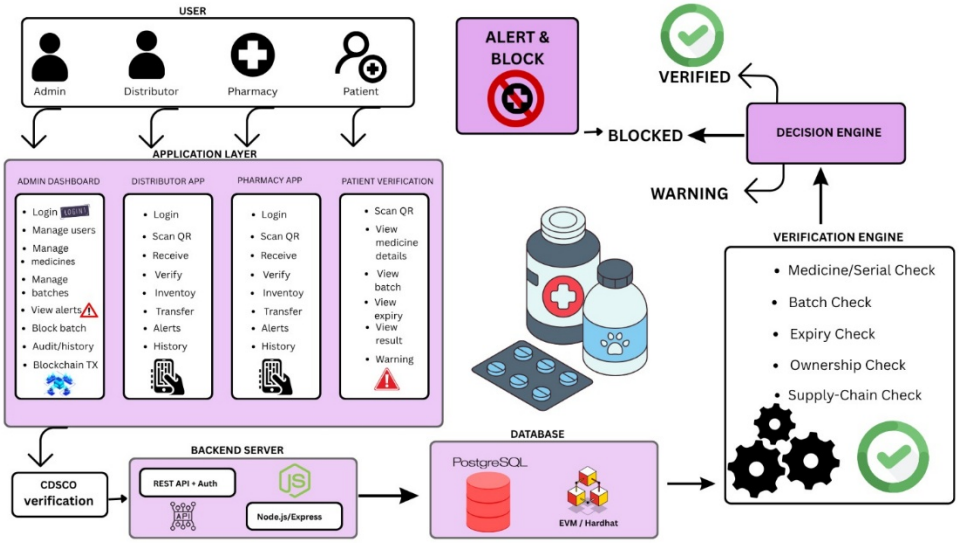


FIG 3.1 ARCHITECTURE DIAGRAM

The first layer of the system includes the users and actors in the pharmaceutical supply chain. The major users of the proposed

verification system are the distributor, pharmacies, and patients. Administrators manage medicines, batches, organizations, regulatory status, blocked batches, and alerts. The manufacturer is an external actor and is out of the scope of the Minimum Viable Product (MVP). The distributor is responsible for receiving, verifying, managing, and transferring medicines. The pharmacy verifies the received medicines and performs dispensing operations, whereas patients can independently scan the QR code of a medicine and view the details of the verification. The administrator manages medicines, batches, organizations, regulatory information, blocked batches, and alerts.

The second layer includes the frontend applications of the system. The Distributor Application involves features, such as login, QR scanning, medicine receiving, verification, inventory, medicine transfer, alert, and history of transactions. Similarly, the Pharmacy Application features include medicine receiving, verification, inventory, dispensing, and alert. The Patient Verification Interface is a simple interface that allows patients to scan a QR code and view the medicine name, batch number, expiry date, regulatory status, and verification result. The Admin Dashboard features include medicine and batch management, distributor and pharmacy management, regulator management, blocked batches, and alerts. Besides, the dashboard provides access to medicines, batches, distributors, pharmacies, regulatory information, blocked batches, alerts, and blockchain transactions.

The third layer includes the Node.js and Express backend API, which serves as the communication layer between the frontends and the system components. The REST APIs feature

authentication, medicine management, batch management, inventory management, QR verification, regulatory information, transfer, dispensing, blockchain, and alerts. The backend is responsible for receiving requests from the different users, validating, and communicating with the corresponding database or blockchain service, and responding with the resultant information to the frontend.

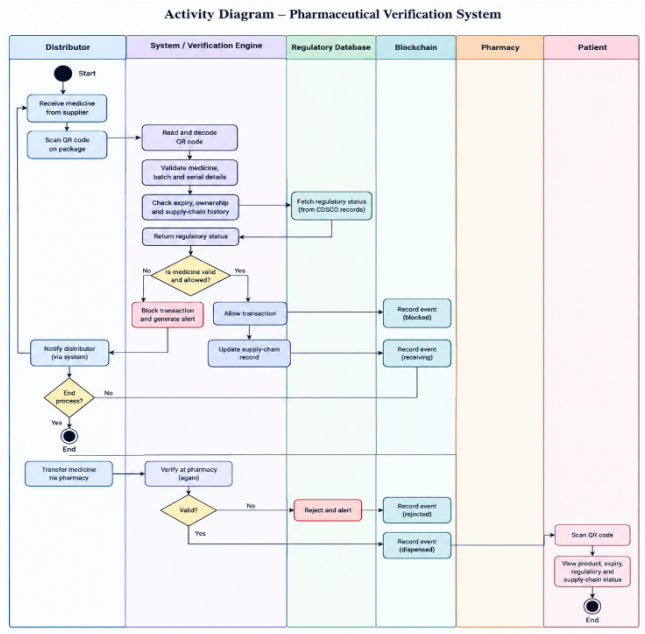
The fourth layer of the system is the Medicine Verification and Enforcement Engine, which is the central business-logic component of the system. For every medicine that is scanned or for which a supply-chain operation is requested, the engine performs medicine identification, expiry validation, regulatory validation, ownership validation, supply-chain validation, blockchain history validation, and dispensing validation. The Decision Engine combines the results of these checks and returns one of the three possible outcomes – verified, warning, or blocked. The system features the major actions of medicine receiving, transferring, dispensing, and patient scanning. Thus, the engine layer ensures that illicit or unauthorized medicines do not pass through the supply chain.

The fifth layer includes the PostgreSQL Operational Database and Regulatory Data Service. PostgreSQL stores the operational information of the system, including the records of users, organizations, medicines, and batches, serials, inventory, regulatory status, transactions, dispensing, and alerts data. The Regulatory Data Service stores the current regulatory information of the medicines or batches, including medicine/batch status, authority, reference number, effective date, reason, source, and last updated information. The regulatory component ensures that regulatory information is

taken into consideration during medicine verification. The sixth layer is the Blockchain Layer. The layer implements a PharmaSupplyChain Smart Contract deployed on an Ethereum-compatible EVM blockchain. For the MVP, the Hardhat local network is used. The smart contract features batch registration, batch receiving, batch transfer, batch blocking, dispensing of medicine, and history of a batch. The information related to batch identifiers, serial identifiers or hash, sender, receiver, transaction type, and timestamp can be stored in the blockchain as proof of certain events. Storing of password, patient personal information, medical records, and other large documents is discouraged in the blockchain.

The seventh layer of the system features Alerts and Audit. The system can feature alerts in instances of counterfeit suspicion, expiry, regulatory block, recall, duplicate serial numbers, ownership mismatch, and supply-chain irregularity. In the MVP version of the system, alerts will be viewable via the dashboards and can be accessed by the respective authorized distributor, pharmacy, and administrator. In the future, the alert system can be extended to include SMS, email, and push notification. The audit also features alerts of all the relevant system activities and blockchain activities.

Verified, Warning, or Blocked. If the medicine satisfies all the test conditions, the requested transaction for the medicine, i.e., receiving, transfer, dispensing, or patient scan will be allowed, and an appropriate supply-chain event will be created. On the other hand, if the medicine is found to be counterfeit, expired, recalled, prohibited, duplicated, or with invalid ownership history or supply history, the corresponding transaction can be blocked, and an alert can be issued.



3.2 SYSTEM REQUIREMENTS

The proposed verification system has the following software requirements. The software should support the frontend using React, backend using Node.js and Express, PostgreSQL, QR processing, and Ethereum compatibility. The system requires hardware, including a computer with Intel Core i5/AMD Ryzen 5, 8 GB RAM, 256 GB SSD. An HD display and the Internet are required. A smartphone with a camera will be used to scan the QR code attached to medicines.

3.2.1 SOFTWARE REQUIREMENTS

SOFTWARE COMPONENT	VERSION	PURPOSE
React.js	18.x	Development of user interfaces
Node.js	20.x	Backend runtime environment
Express.js	4.x	Development of REST APIs
PostgreSQL	16.x	Storage of operational and regulatory data

ethers.js	6.x	Backend-Blockchain Communication

3.2.2 HARDWARE REQUIREMENTS

The proposed system requires a computer with an Intel Core i5/AMD Ryzen 5 processor, 8 GB RAM, and 256 GB SSD storage for application development, database management, and blockchain testing. A standard HD display and a stable Internet connection are required for development and system communication. A camera-enabled smartphone is used for scanning the QR codes attached to medicines during verification.

3.2.3 DATASET REQUIREMENTS

The proposed verification system does not have a conventional machine learning dataset. Medicine data will be provided via API, and regulatory will be provided by CDSCO. Medicine and regulatory information will be provided by CDSCO. Thus, the system can be updated with the latest medicine and regulatory information. Besides, APIs can be used to provide medicine/batch data. CDSCO data and alerts can be used to update the regulatory status of medicines, such as prohibited, recalled, restricted, or permitted. The system includes additional samples of medicine, such as valid, counterfeit, expired, duplicate, recalled, and unauthorized for testing purposes.

CHAPTER 4
PROJECT DESCRIPTION

4.1 METHODOLOGIES

The suggested system employs the following hybrid methodology, which consists of QR-based medicine identification, API-based data authentication, regulatory validation based on CDSCO, database management, and blockchain technology. Each individual medicine is verified for its authenticity, validity, regulation, ownership, and chain of supply. The legitimate transactions are recorded in blockchain through the smart contracts, and the illegitimate ones (expired, prohibited, or recalled) are denied, and appropriate alarms are raised.

4.2 MODULES

Some of the modules that make up the proposed system are User Authentication and Role Management, Medicine Registration and QR Code Validation, Distributor and Pharmacy Management, Inventory and Supply Chain Management, Regulatory Verification, Blockchain and Smart Contract Management, Alert and Notification Management, and Patient Verification. These modules combine to perform functions such as user authentication, medicine validation, tracking of ownership and transactions, regulatory validation by means of API and CDSCO data, storing key events in the blockchain, preventing unsafe medicines, and allowing patients to validate medicines.

4.2.1 USER AUTHENTICATION AND ROLE MANAGEMENT

The system manages the process of secure user authentication and role-based access for distributors, pharmacies, patients, and administrators. The users will authenticate themselves via their credentials, and then access to functions within the system will be granted based on their roles. This will ensure that only authorized individuals will be able to perform actions like receiving, moving, or dispensing drugs.

4.2.2 MEDICINE REGISTRATION AND QR VERIFICATION

In this module, registration of the medicines and the batches is regulated, as well as QR recognition is done. Each medicine or batch has a single QR code that will provide you with the information about the medicine. After identifying the information such as identification of the medicine, the batch or the serial number, the expiration date, etc., verification will be shown.

4.2.3 DISTRIBUTOR AND PHARMACY MANAGEMENT

This module deals with the management of the activities of the distributor and the pharmacy in the pharmaceutical chain. The distributor may receive and distribute the medicine, whereas the pharmacy may receive the medicine, store it, and dispense the medicine that is approved.

4.2.4 INVENTORY AND SUPPLY-CHAIN TRACKING

This module is used to record the inventory level and monitor the movement of the medicine through the supply chain. The transaction of receipt, transfers, and dispensations must be approved before entering the data into the system. This will assist in tracking the ownership of the medicine and monitor the movement of medicines from distributor to pharmacies and ultimately to patients.

4.5 REGULATORY VERIFICATION

This module is responsible for verifying the regulatory status of the medicines and batches using the data derived from APIs and CDSCO. The medicine can be categorized as permitted, restricted, recalled, prohibited, expired, and unknown. In the case where the medicine or batch falls under any regulatory restriction or recall, then the status will be updated.

4.6 BLOCKCHAIN AND SMART CONTRACT MANAGEMENT

In this module, a blockchain system compatible with Ethereum is used for maintaining a tamper-proof record of key events of the supply chain. Events like batch registration, receipt, transfer, blockage, and dispensation are logged using the smart contract. The smart contract will further enforce predefined policies and will not allow transactions of invalid, expired, or recalled medicines.

4.7 ALERT AND AUDIT MANAGEMENT

This module creates an alert in case any suspicious or incorrect situation is identified during verification or supply chain management. These alerts can be created in case there is a suspicion of counterfeits, out-of-date medicines, recalled batches, duplicate codes, ownership discrepancies, or medicines that should not be there. This module creates an audit trail of important actions.

4.8 PATIENT VERIFICATION

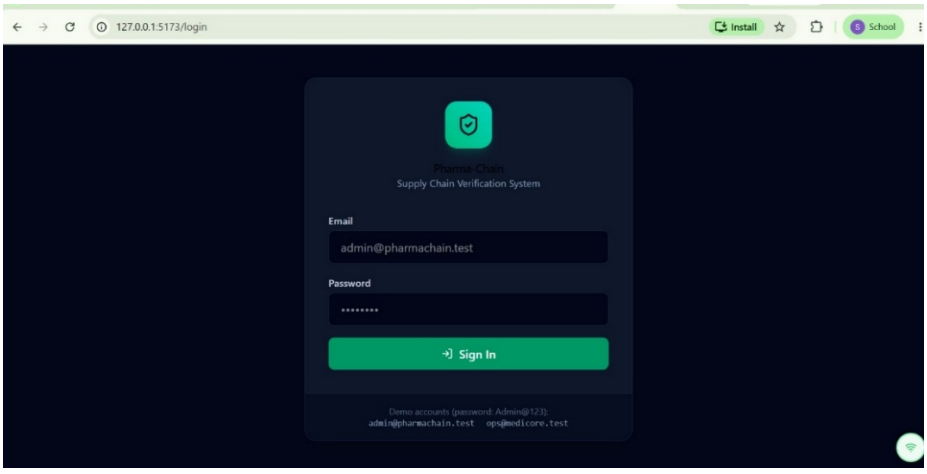
The patient verification module enables patients to independently check medicines by scanning the QR code that is present on the medicine packaging. The display screen shows the required information related to the medicine name, batch number, expiry information, regulatory status, and verification result. This will give the patients more visibility into the process.

CHAPTER 5
IMPLEMENTATION AND RESULT DISCUSSION

5.1 IMPLEMENTATION RESULT

5.1.1 Implementation of User Authentication and Role-Based Access

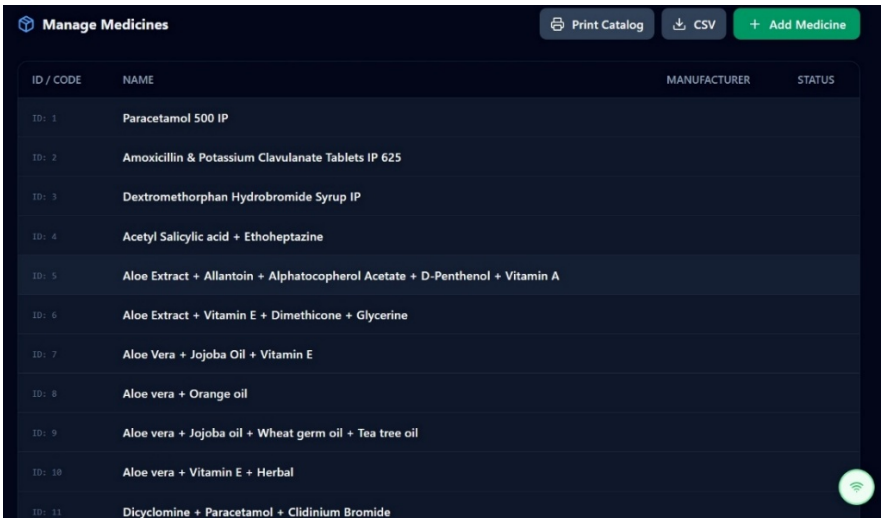
This work introduces an authentication technique in order to ensure the security of the involved users in the process of pharmaceutical supply chain management. In the implementation, there are different roles for distributors, pharmacies, patients, and administrators. Prior to any operation carried out by the users, each of the users is required to be authenticated in order to use the service provided by the system. Role-based access control is also provided so that only authorized users are allowed to perform some particular operation. Distributors and pharmacists are provided with operations related to product verification, managing inventory, receipt of medicine, and transfer of medicine. Patients are provided with a simple interface where they are able to scan the QR code of a medicine and get its verification result. Administrators have additional operations regarding management of medicine record, batch, user, transaction, and alert



5.1.2 Implementation of Medicine and Batch Management

In the medicine management module, there is basic information which enables the identification of pharmaceutical products. There is medicine name, manufacturer,

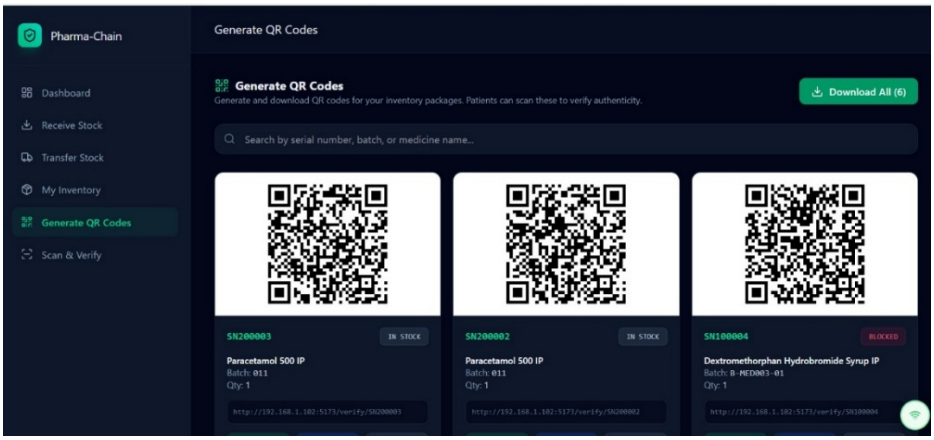
dosage details and information related to the identity of the medicine. Batch details are stored in a separate table since different batches of the same medicine can have different manufacturing and expiry dates. Batches are distinguished through batch ID numbers. Medicine units can also be identified by unique serial numbers. There are database links between medicines, batches, serial numbers, inventory information and organizations involved. Using this arrangement, it is possible for the system to link medicine information using its QR code.



5.1.3 Implementation of QR/Barcode-Based Medicine Verification

QR/barcodes verification module serves as the main interface between the physical medicine packaging and our suggested software-based system. The QR code which corresponds to the certain medicine stores all the necessary data about the

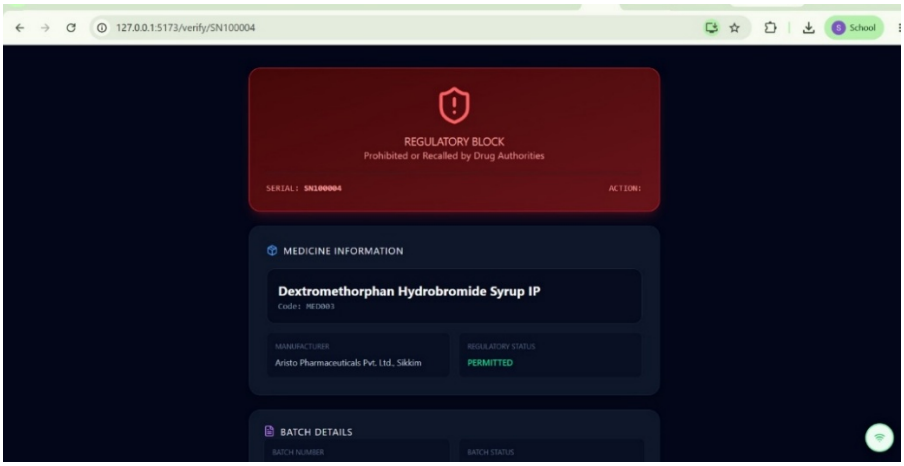
medicine, its batch number, and serial number. As soon as a distributor, pharmacy, or patient scan this code, the application sends all this data to the server using a REST API. The server checks the provided information and gets all the necessary records from the database. The verification of the medicine is not dependent on the availability of valid QR code, instead, there will be multiple stages of the verification of this medicine, including serial number validity check, batch verification, checking the expiration of the medicine, verifying the ownership information and supply chain transactions. Thus, the whole process of the verification includes such stages as: QR code scanning, product identification, database verification, blockchain verification, decision-making, and result displaying.



5.1.4 Implementation of the Medicine Verification Engine

The Medicine Verification and Enforcement Engine functions as the primary processing element in the proposed solution architecture. The medicine identification code, batch identification code, serial number, organization of the requesting user, and requested operation constitute the input for the engine. The engine processes the inputs using a series of verification operations. Firstly, the Medicine and Serial Number Verification module verifies the existence of the medicine and

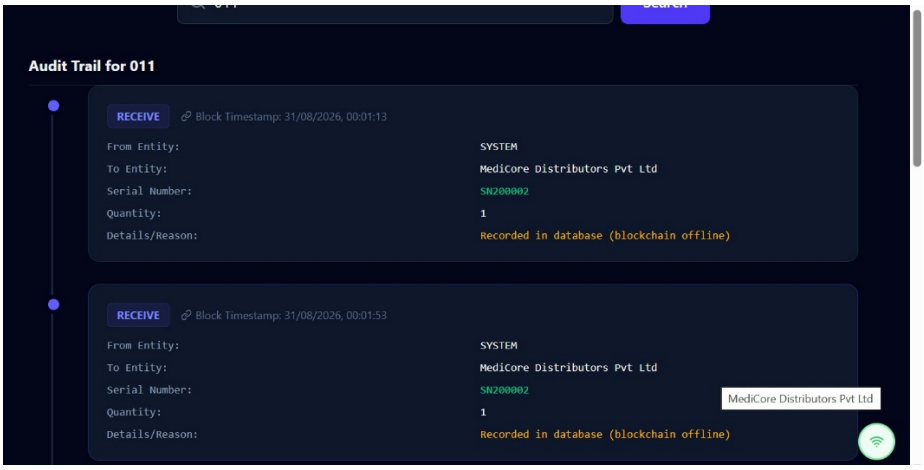
the serial number in the system. Secondly, the Batch Verification module verifies the existence of the specific batch in the system. Thirdly, the Expiry Date Verification module validates the medicine expiry date against the present date. Finally, the Ownership Verification module verifies the ownership of the medicine by the specified organization. Finally, the duplicate verification checks for duplicated or contradictory uses of the serial number. The last step is the Decision Engine, which aggregates the results from all other checks and determines the final verification result. As a result of this process, the system will output one of the following three main results: VERIFIED—the medicine meets the verification requirements; WARNING – the medicine includes some suspicious condition; BLOCKED – the medicine does not meet the verification requirement, and further processing is blocked. This centralized decision approach guarantees consistent application of the verification algorithm to different types of verifications.



5.1.5 Implementation of Blockchain-Based Supply-Chain Tracking

Blockchain technology is employed for recording tamper-proof records of transactions within the pharmaceutical supply chain.

Ethereum-compatible EVM is employed for software development and testing, and the Solidity smart contract manages the supply chain processes. These processes include batch registration, batch receiving, batch transfer, medicine dispensing, batch blocking, and retrieving transaction history of batches. In case of supply chain events, the communication between the backend and the blockchain takes place using the blockchain service with the help of ethers.js. In this case, the smart contract function gets invoked, and the transaction hash is stored at the backend. The transaction process then follows the order of application request, backend API, blockchain service, smart contract, blockchain network, transaction confirmation, and transaction hash storing. The blockchain system saves the necessary transaction data instead of application or patient data.



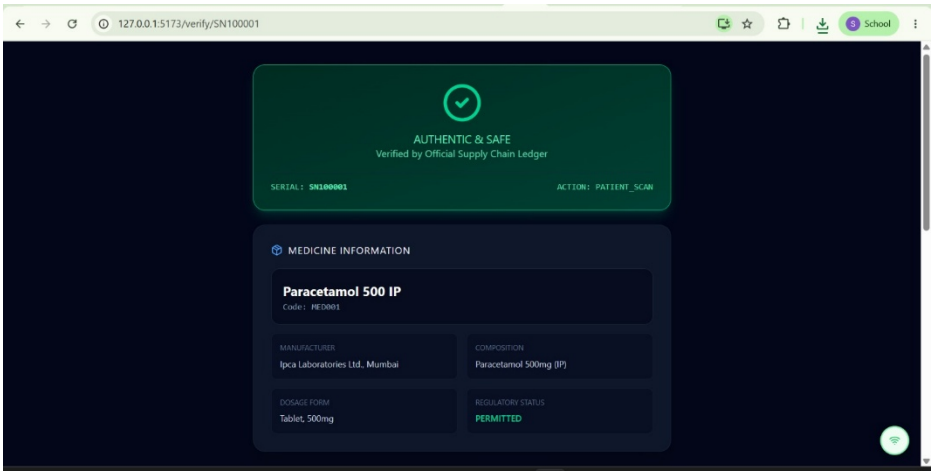
5.1.6 Implementation of Distributor and Pharmacy Operations

The functionality provided by the distributor module includes receiving pharmaceutical batches and sending them to the appropriate pharmacies. Upon receipt of a batch by the distributor, the system checks the details of the batch and only then proceeds to record the transaction. When transferring a

batch from a distributor to a pharmacy, the system will check the current ownership of the batch and record the transfer between the two entities. The transaction is also recorded on the blockchain. The pharmacy module offers functionality similar to that of the distributor module for receiving medicines from the distributor. Prior to the addition of the batch to inventory, a verification process may be carried out. Dispensing of the medicine is recorded by the system along with its corresponding transaction.

5.1.7 Implementation of Patient Verification

The patient interface is designed as a simplified verification mechanism. A patient does not require access to internal inventory, ownership, or administrative information. After scanning the QR code, the system retrieves the relevant medicine information and performs the verification process. The patient interface displays the resulting status along with appropriate product information. For a valid medicine, the system can display the medicine name, batch identifier, expiry information, and a VERIFIED status. If the medicine fails an important verification condition, the interface can instead display a BLOCKED status with an appropriate reason, such as an invalid, expired, or suspicious product. This provides patients with a simple mechanism for checking the validity of medicines already circulating in the market without exposing sensitive internal supply-chain information.



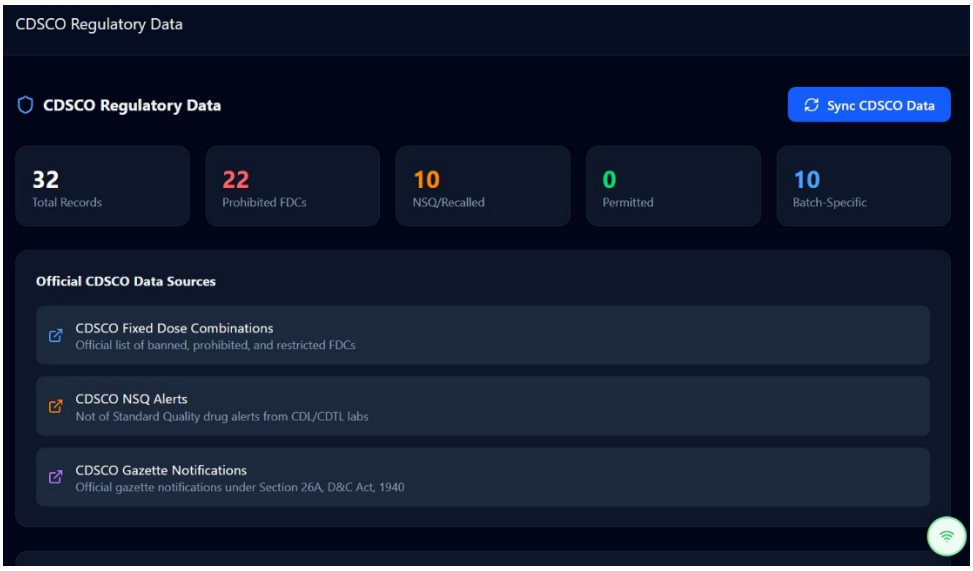
5.1.7 Implementation of Alert and Enforcement Mechanism

The alert engine is to be utilized in order to notify the relevant users about verification failure and other suspicious cases. Some of the suspicious situations that the system can alert on include expiration of medication, serial number validation failure, duplicate serial number, wrong ownership, validation of supply chain sequence, suspicious medication, and blocked batches. Whenever the Verification Engine outputs a BLOCKED response, the relevant alert will be saved in the database and presented via the proper application interface. Supply chain users are to be prevented from further processing of the respective medicine by the blocking process. The administrator may view the alert and examine the related medication, batch, and transactions.

5.1.8 Implementation of Database Management

PostgreSQL is used as the primary off-chain database for storing application and operational information. The database is organized into multiple related entities, including users, organizations, medicines, batches, serial numbers, inventory, transactions, dispensing records, alerts, and audit logs. The database stores detailed application information that would be

inefficient or inappropriate to store directly on the blockchain. Blockchain transaction hashes and relevant references can be associated with database records to establish a connection between the off-chain application data and on-chain transaction history. This hybrid storage model provides efficient data retrieval through PostgreSQL while using blockchain for tamper-resistant transaction tracking



5.1.9 Implementation of Audit and Security Mechanisms

The security features are implemented in the backend so as to safeguard the system against any unauthorized entry and manipulation. The JWT-based authentication technique is used in session authorization, and the role-based access control is used in controlling the operations based on the roles of the user. Input validation is implemented on the incoming API request in order to avoid invalid input from accessing the system. Important actions made by the users and the system are logged in audit logs. Smart-contract access control mechanisms are also implemented to secure the blockchain actions. Thus, the system implements two different kinds of records: audit logs in the database and the blockchain records for transaction history in the supply chain.

5.2 DISCUSSION

The implementation confirms that blockchain technology can be combined with a classical pharmaceutical management system to increase the traceability and verification of the process. Instead of implementing blockchain as the main database for the application, the proposed system suggests the use of a hybrid structure where PostgreSQL will store all application-related data, and the blockchain will store the relevant events from the supply chain. It is possible to save space on-chain and keep the transactional records intact. Among the significant benefits of the proposed solution is that it is possible to verify the medicine when it is already in the market. Any of the participants can perform verification of the medicine by scanning the QR code of the product. In this way, several parameters are evaluated instead of one, increasing the possibility to detect invalid, expired, duplicated, or suspicious medicines.

As shown in the experimentation, the system is able to detect various issues such as expired medicines, invalid serial numbers, duplicate serial numbers, owner mismatch, and supply chain transaction abnormalities. The inclusion of a central verification engine is helpful in ensuring that these verifications are done consistently depending on the user types. The classification results of VERIFIED, WARNING, and BLOCKED help in clearly communicating the result of the verification.

The inclusion of blockchain as a component of the system helps in adding an extra level of trust, as the supply chain events are stored in the form of tamper-proof transactions upon confirmation. This is particularly helpful in storing the history of the movement of medicine from one organization to another.

This system is a prototype, and it is mostly tested within a controlled development environment. The blockchain, database, and users involved are all restricted to the experimental scenario. In order to use this system on a large scale, there will have to be an evaluation done using much more substantial amounts of medicines, batches, transactions, and users simultaneously. Evaluation will be needed in order to determine transaction latency, API response time, database performance, blockchain throughput, and scalability.

Generally, the results of this implementation show that the system being designed is a viable framework for medicine verification, traceability of the supply chain, suspicious medicine identification, and medicine monitoring after they go to market based on QR code identification, central verification, and blockchain-verified transaction history.

CHAPTER 6
CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

In this regard, the proposed system provides a pharmaceutical verification and supply chain tracking system that is based on the use of a blockchain technology. The system provides QR/barcode-based medicine verification, a centralized engine, a PostgreSQL database, and an Ethereum blockchain platform. This combination allows pharmaceutical products to have a digital record while keeping a tamper-proof history of essential transactions of pharmaceutical products' supply chain. The implemented system has different user interfaces for distributors, pharmacies, patients, and administrators. The distributor and pharmacy are able to verify medicines, maintain inventory, receive products, and make transfers, whereas the patient is able to scan the medicine and get verification information about that product using a simplified interface. The administrator will manage medicine information, transactions, alerts, and records of the system. Role-based access control has been provided so that participants are only allowed to carry out the operations relevant to them. The multi-level verification has been implemented using the Medicine Verification and Enforcement Engine. The engine checks the identity of the medicine, batch information, validity of the serial number, expiry information, ownership, and suspicious activities.

From the above tests, the system produces outcomes which can be either VERIFIED, WARNING, or BLOCKED. This strategy ensures a more thorough verification process compared to only the presence of a QR code or product identifier. Blockchain is

utilized as a trustworthy traceability tool for capturing essential events related to the product in the supply chain such as batch registration, receipt, transfer, blocking, and dispensing. Using a smart contract helps in generating a tamper-proof history of transactions that can be referred back to in the subsequent verification process. On the other hand, all details about the application are stored in PostgreSQL, hence creating a hybrid design where full application data is not stored in the blockchain. The implementation and testing process has proved that the suggested system can detect essential pharmaceutical conditions including expired medicines, invalid serial number, duplicate serial number, inconsistency in ownership, and unusual transaction in the supply chain. The alerts and enforcement system will allow problems to be notified to the respective users and the blocked medicines to be prevented from proceeding in the processing.

This system helps in developing an infrastructure to detect any suspicious products which are being distributed within the market. This system, though at a prototype stage under controlled conditions, could definitely be scaled up to monitor and manage larger quantities of pharmaceutical products.

6.2 FUTURE WORK

Future developments of the suggested solution can concentrate on the improvement of its scalability, automation, detection capabilities, and practical implementation. One of the possible development directions is the use of an automated regulation data synchronization with the authorities providing pharmaceutical regulations. In place of manually maintained regulation data, the system can perform periodical retrieval of

new regulatory notifications, drug alerts, recalls, and forbidden product information; detect the changes in the obtained data; normalize the information and update the regulation database. Thus, the verification engine will be able to work with more fresh information when checking medicines that are already in circulation. Another possible direction is the incorporation of the manufacturer-based verification into the supply chain structure. In future versions, it will be possible to allow verified manufacturers to register the product information and create authentic medicine and batch records before products become available for distribution. Such approach will increase the connection between the physical product and its virtual identity and may facilitate counterfeit and suspicious products detection. Machine learning techniques can also be used for improving detection capabilities.

Deployment on the permissioned or consortium blockchain platform can also be studied to find out whether it is better for cooperation among the organizations within the pharmaceutical supply chain. Finally, the suggested system can be evolved into a mobile and national-level pharmaceutical verification platform. The mobile app can offer easy QR-based verification for distributors, pharmacists, and patients. Additional means of notification such as emails or text messages can also be used for important notifications. Privacy-preserving methods can also be studied to protect patient information from any leakage but at the same time let the right parties verify pharmaceuticals. In general, further development will concentrate on transforming the existing prototype into a scalable and intelligent verification platform for pharmaceutical products.

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PLAGIARISM REPORT

